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Color Measurement and Visual Matching

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This report consolidates the Project 2 measurement workflow, numerical results, visual comparisons, and spectral plots. Two paint samples (3.1 and 3.2), their photographed/displayed reproductions, and visually matched display colors were measured with a ColorMunki and ArgyllCMS.

The measurements include spectral data from 380 to 730 nm in 10 nm steps, CIE XYZ tristimulus values, and CIELAB values. Differences in CIELAB are calculated relative to each physical sample.

```
projectDir = fileparts(fileparts(mfilename('fullpath')));
projectsDir = fileparts(projectDir);
dataDir = fullfile(projectDir, 'data');
resultsDir = fullfile(projectDir, 'results');

colorimetry = load(fullfile(dataDir, 'patchColorimetryData.mat'));
spectra = load(fullfile(dataDir, 'patchSpectralData.mat'));
wavelengths = 380:10:730;

assert(numel(wavelengths) == numel(spectra.p31_real), ...
    'The wavelength axis does not match the saved spectral measurements.');
```

Experimental comparison

The first photograph shows the physical samples beside their photographed reproductions from Project 1. The second shows the physical samples beside display colors adjusted by visual matching.

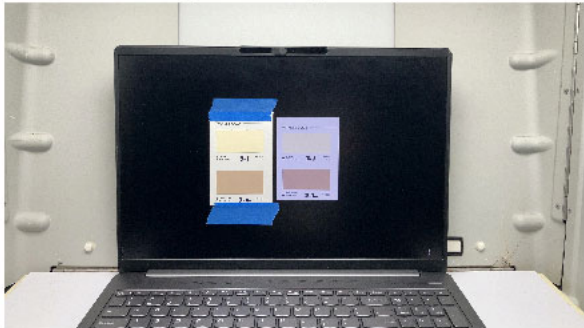
```
realVsImagedFile = fullfile(projectsDir, '01-color-problem', 'results', ...
    'comparison_01.JPG');
realVsMatchedFile = fullfile(dataDir, 'colormunki_data', 'matched', ...
    '3_real_vs_matching.jpg');

comparisonFigure = figure('Color', 'w', 'Name', ...
    'Physical, imaged, and visually matched samples');
comparisonLayout = tiledlayout(comparisonFigure, 1, 2, ...
    'TileSpacing', 'compact', 'Padding', 'compact');

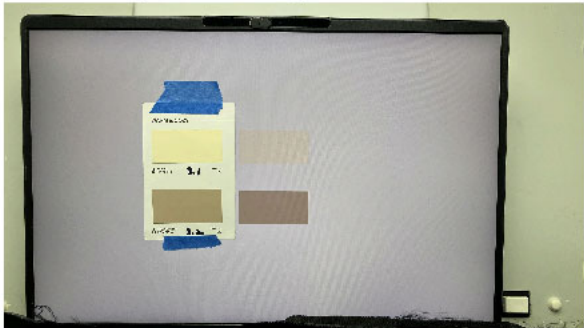
nexttile(comparisonLayout);
imshow(imread(realVsImagedFile));
title('Physical vs. imaged samples');

nexttile(comparisonLayout);
imshow(imread(realVsMatchedFile));
title('Physical vs. visually matched samples');
```

Physical vs. imaged samples



Physical vs. visually matched samples



Measured CIE XYZ and CIELAB values

The saved data arrays contain an original measurement identifier in column 1; columns 2 through 7 contain X, Y, Z, L*, a*, and b*. The fixed-width layout preserves the presentation used in the original project report.

```
fprintf('\n%s\n', "Measured XYZ and Lab values");

% Print the table for patch 3.1.
fprintf('\n%8s %8s %8s %8s %8s %8s\n', ...
    '', '', '', 'patch 3.1', '', '', '');
fprintf('%5s %8s %8s %8s %8s %8s\n', ...
    '', 'X', 'Y', 'Z', 'L', 'a', 'b');
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'real', colorimetry.XYZLabsreal(1, 2:end));
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'imaged', colorimetry.XYZLabsimaged(1, 2:end));
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'matching', colorimetry.XYZLabsmatching(1, 2:end));

% Print the table for patch 3.2.
fprintf('\n%8s %8s %8s %8s %8s %8s\n', ...
    '', '', '', 'patch 3.2', '', '', '');
fprintf('%5s %8s %8s %8s %8s %8s\n', ...
    '', 'X', 'Y', 'Z', 'L', 'a', 'b');
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'real', colorimetry.XYZLabsreal(2, 2:end));
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'imaged', colorimetry.XYZLabsimaged(2, 2:end));
fprintf('%8s %8.4f %8.4f %8.4f %8.4f %8.4f %8.4f\n', ...
    'matching', colorimetry.XYZLabsmatching(2, 2:end));
```

Measured XYZ and Lab values						
patch 3.1						
	X	Y	Z	L	a	b
real	42.2036	41.3758	25.3596	70.4384	7.0533	14.0496
imaged	44.1598	43.6064	28.4608	71.9646	6.2521	11.3891
matching	85.2686	87.3024	57.3942	94.8664	2.0563	13.9269
patch 3.2						
	X	Y	Z	L	a	b
real	32.3292	34.2553	31.9201	65.1647	-2.4888	-5.8023
imaged	63.9017	65.2199	49.3994	84.5969	2.3239	4.8643
matching	44.0165	42.7388	21.1155	71.3773	8.3661	23.6626

CIELAB component differences

Differences are calculated as comparison minus physical reference. Positive delta L* indicates lighter, positive delta a* indicates redder, and positive delta b* indicates yellower. Negative delta a* indicates greener, while negative delta b* indicates bluer.

```
physicalLab = colorimetry.XYZLabsreal(:, 5:7);
imagedLab = colorimetry.XYZLabsimaged(:, 5:7);
matchedLab = colorimetry.XYZLabsmatching(:, 5:7);

deltaImaged = imagedLab - physicalLab;
deltaMatched = matchedLab - physicalLab;

differenceValues = [
    deltaImaged(1, :)
    deltaMatched(1, :)
    deltaImaged(2, :)
    deltaMatched(2, :)
];

Patch = ["3.1"; "3.1"; "3.2"; "3.2"];
Comparison = ["Imaged - physical"; "Matched - physical"; ...
    "Imaged - physical"; "Matched - physical"];
LabIndication = [
    "Slightly lighter, greener, and bluer"
    "Much lighter and greener; nearly unchanged yellow-blue component"
    "Much lighter, redder, and yellower"
    "Lighter, redder, and much yellower"
];
VisualObservation = [
    "Appeared bluer and lighter, with little red-green change"
    "Appeared much lighter and greener, with a small blue shift"
    "Appeared slightly redder and yellower, but darker rather than lighter"
    "Appeared yellower and slightly redder, but darker than the physical patch"
];

differenceTable = table(Patch, Comparison, differenceValues(:, 1), ...
    differenceValues(:, 2), differenceValues(:, 3), LabIndication, ...
    VisualObservation, 'VariableNames', {'Patch', 'Comparison', ...
    'DeltaLStar', 'DeltaaStar', 'DeltabStar', 'LabIndication', ...
    'VisualObservation'});

disp('CIELAB differences relative to each physical patch');
disp(differenceTable);
```

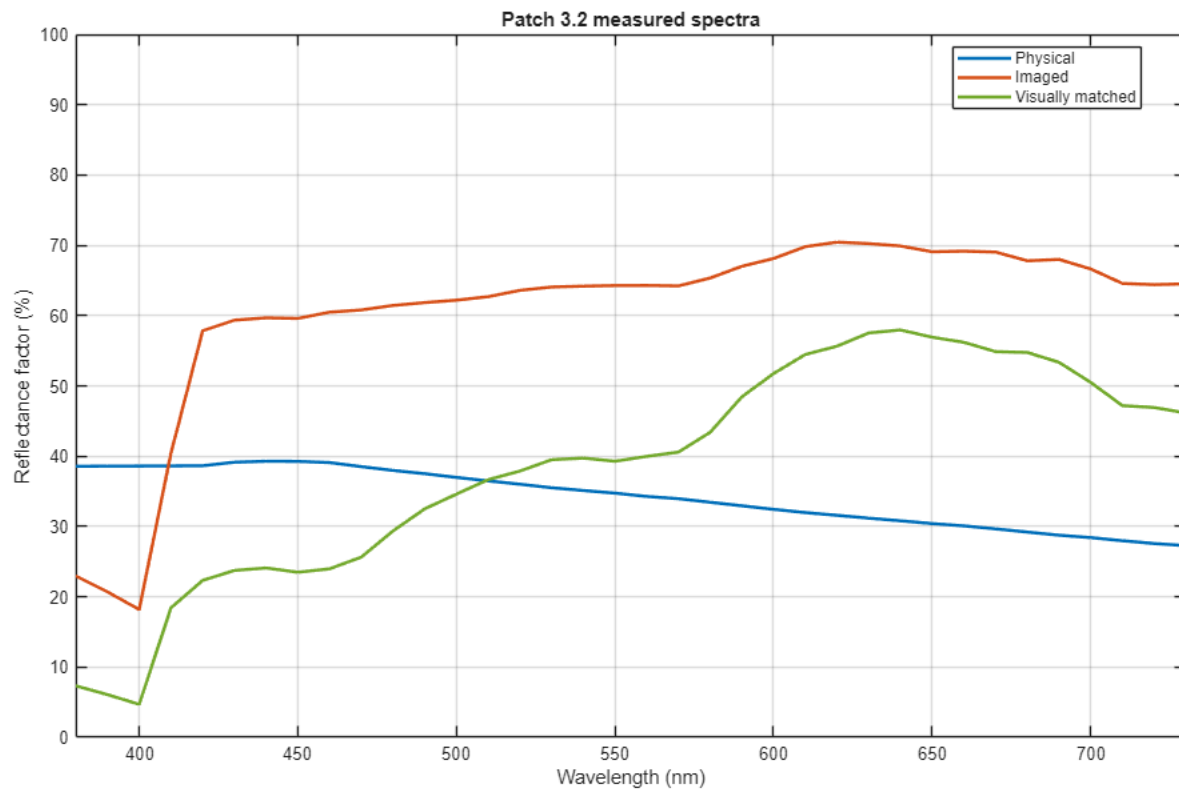
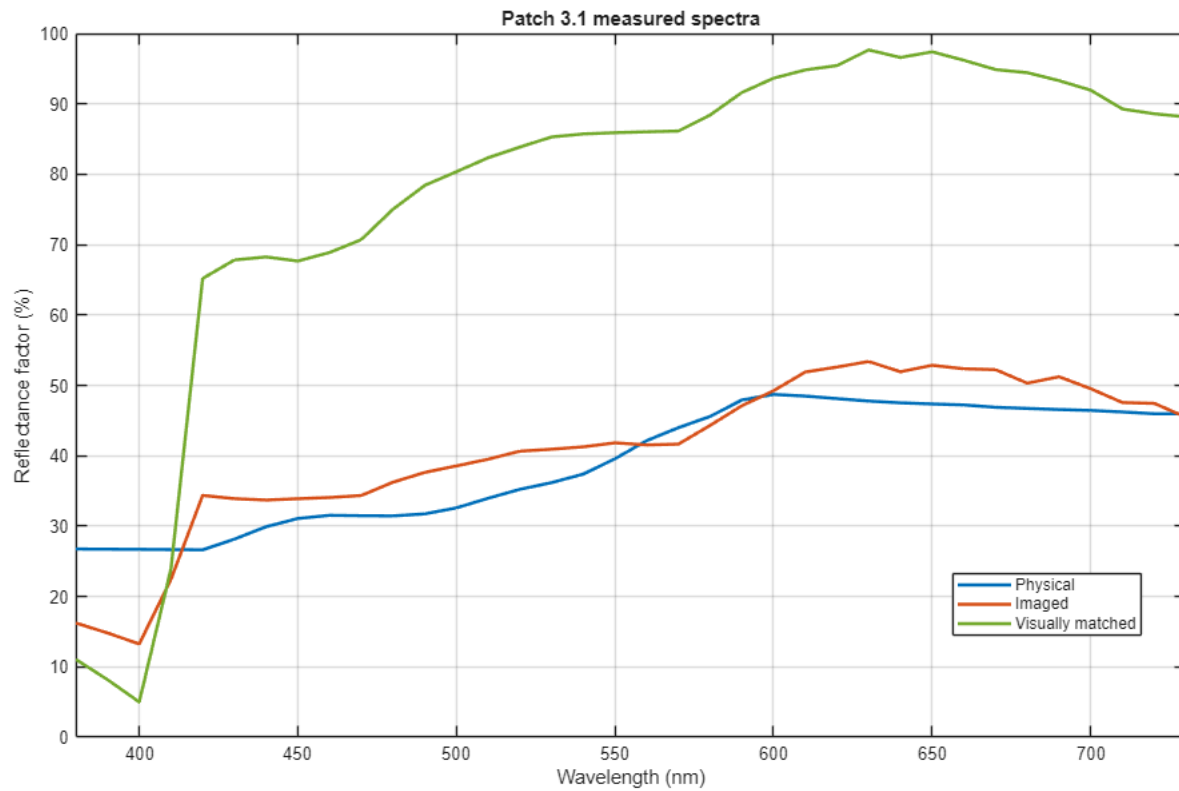
CIELAB differences relative to each physical patch						
Patch	Comparison	DeltaLStar	DeltaaStar	DeltabStar	LabIndication	
"3.1"	"Imaged - physical"	1.5262	-0.80129	-2.6605	"Slightly lighter, greener, and bluer"	"Appeared bluer
"3.1"	"Matched - physical"	24.428	-4.9971	-0.12269	"Much lighter and greener; nearly unchanged yellow-blue component"	"Appeared much
"3.2"	"Imaged - physical"	19.432	4.8127	10.667	"Much lighter, redder, and yellower"	"Appeared sligh
"3.2"	"Matched - physical"	6.2126	10.855	29.465	"Lighter, redder, and much yellower"	"Appeared yello

Measured reflectance spectra

The plots use a consistent color mapping in both figures: physical is blue, imaged is orange, and visually matched is green. The older plotting script labeled the physical and imaged curves in the opposite order; the data were correct, but the legend mapping was not.

```
plotPatchSpectra(wavelengths, spectra.p31_real, spectra.p31_imaged, ...
    spectra.p31_matching, 'Patch 3.1', ...
    fullfile(resultsDir, 'patch3.1_spectral_plot.png'));

plotPatchSpectra(wavelengths, spectra.p32_real, spectra.p32_imaged, ...
    spectra.p32_matching, 'Patch 3.2', ...
    fullfile(resultsDir, 'patch3.2_spectral_plot.png'));
```



Summary

Patch 3.1's imaged measurement remained relatively close to the physical sample in CIELAB, while its visually matched measurement was substantially lighter. For patch 3.2, both comparison conditions shifted toward positive a^* and b^* , indicating redder and yellower measurements. The visual judgments generally agreed with the hue directions but did not always agree with the measured lightness direction, illustrating the difference between instrument measurements and appearance in a camera-display viewing workflow.

```

fprintf('\nSummary of component differences (comparison minus physical):\n');
fprintf('Patch 3.1 imaged:  dL*=%7.4f, da*=%7.4f, db*=%7.4f\n', deltaImaged(1, :));
fprintf('Patch 3.1 matched: dL*=%7.4f, da*=%7.4f, db*=%7.4f\n', deltaMatched(1, :));
fprintf('Patch 3.2 imaged:  dL*=%7.4f, da*=%7.4f, db*=%7.4f\n', deltaImaged(2, :));
fprintf('Patch 3.2 matched: dL*=%7.4f, da*=%7.4f, db*=%7.4f\n', deltaMatched(2, :));

function plotPatchSpectra(wavelengths, physical, imaged, matched, ...
    patchName, outputFile)
%PLOTPATCHSPECTRA Plot and export physical, imaged, and matched spectra.

figureHandle = figure('Color', 'w', 'Name', patchName);
hold on;
plot(wavelengths, physical, 'Color', '#0072BD', 'LineWidth', 1.8);
plot(wavelengths, imaged, 'Color', '#D95319', 'LineWidth', 1.8);
plot(wavelengths, matched, 'Color', '#77AC30', 'LineWidth', 1.8);
hold off;

xlim([380 730]);
ylim([0 100]);
xticks(400:50:700);
yticks(0:10:100);
grid on;
box on;
xlabel('Wavelength (nm)');
ylabel('Reflectance factor (%)');
title([patchName ' measured spectra']);
legend({'Physical', 'Imaged', 'Visually matched'}, ...
    'Location', 'best');

exportgraphics(figureHandle, outputFile, 'Resolution', 160);
end

```

```

Summary of component differences (comparison minus physical):
Patch 3.1 imaged:  dL*= 1.5262, da*=-0.8013, db*=-2.6605
Patch 3.1 matched: dL*=24.4280, da*=-4.9971, db*=-0.1227
Patch 3.2 imaged:  dL*=19.4322, da*= 4.8127, db*=10.6666
Patch 3.2 matched: dL*= 6.2126, da*=10.8549, db*=29.4649

```